

Midterm Review



Midterm outline

- ❑ Introduction
- ❑ Pseudocode
- ❑ Insertion sort
- ❑ Mergesort
- ❑ Quicksort
- ❑ Analysis of algorithms (O , Ω , Θ).
- ❑ Substitution method
- ❑ Master method
- ❑ Recurrence trees
- ❑ The hiring problem

Midterm outline

- ❑ Lectures 1-6 (including)
- ❑ Summation formulas and properties
- ❑ Sorting algorithms
- ❑ Recurrence equations

Midterm review

- Very useful properties of summation:

Arithmetic series

The summation

$$\sum_{k=1}^n k = 1 + 2 + \cdots + n ,$$

is an *arithmetic series* and has the value

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{A.1}$$

$$= \Theta(n^2) . \tag{A.2}$$

Midterm review

Geometric series

For real $x \neq 1$, the summation

$$\sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$$

is a *geometric series* and has the value

$$\sum_{k=0}^n x^k = \frac{x^{n+1} - 1}{x - 1} . \tag{A.6}$$

Midterm review

The infinite decreasing geometric series occurs when the summation is infinite and $|x| < 1$:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x} . \quad (\text{A.7})$$

If we assume that $0^0 = 1$, these formulas apply even when $x = 0$.

Harmonic series

For positive integers n , the n th *harmonic number* is

$$\begin{aligned} H_n &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} \\ &= \sum_{k=1}^n \frac{1}{k} \end{aligned} \quad (\text{A.8})$$

$$= \ln n + O(1) . \quad (\text{A.9})$$

Assignment 1 – Exercise 1

- a. $T(n) = 2T(n/3) + \Theta(n)$ has solution $T(n) = \Theta(n)$. (17.5%)

Let c be the constant in the $\Theta(n)$ term. To prove the upper bound, we guess that $T(n) \leq dn$. $T\left(\frac{n}{3}\right) = \frac{dn}{3}$, where d is a constant of our choice:

$$\begin{aligned} T(n) &= 2T\left(\frac{n}{3}\right) + cn \leq \frac{2dn}{3} + cn \\ &= n\left(\frac{2d}{3} + c\right) \leq dn \\ &= \frac{2d}{3} + c \leq d \\ &= d \geq 3c. \end{aligned}$$

The lower bound could be proved by changing the sign of the inequality, however the constants must be chosen carefully. For the upper bound, the constant d needs to be 3 times $\geq c$. For the lower bound $d \leq 3c$, $d = c$ works.

Assignment 1 – Exercise 1

b. $T(n) = 4T(n/2) + \Theta(n)$ has solution $T(n) = \Theta(n^2)$. (17.5%)

Let c be the constant in the $\Theta(n)$ term. We guess that $T(n) \leq dn^2 - en$. Where d and e are constants that we will choose. To show the upper bound we have:

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + cn \\ &= 4\left(d\left(\frac{n}{2}\right)^2 - \frac{en}{2}\right) + cn \\ &= dn^2 - 2en + cn \\ &\leq dn^2 - en \\ &= -en \leq -cn \\ &= e \geq c \end{aligned}$$

Assignment 1 – Exercise 2

a. $T(n) = T(n/5) + \Theta(n^3)$. (17.5%)

The recurrence has the following recursion tree:

	Level	Tree	Cost per level
	0	cn^3	cn^3
	1	$c\left(\frac{n}{5}\right)^3$	$cn^3\left(\frac{1}{125}\right)$
	2	$c\left(\frac{n}{25}\right)^3$	$cn^3\left(\frac{1}{125}\right)^2$
	3	$c\left(\frac{n}{125}\right)^3$	$cn^3\left(\frac{1}{125}\right)^3$
		$\vdots$	
	i	$c\left(\frac{n}{5^i}\right)^3$	$cn^3\left(\frac{1}{125}\right)^i$



$\log_5 n + 1$

Assignment 1 – Exercise 2

In total there are $\log_5 n + 1$ levels in the recursion tree. If we sum the costs, we get:

$$\text{Total: } cn^3 \sum_{i=0}^{\log_5 n} \left(\frac{1}{125}\right)^i < cn^3 \sum_{i=0}^{\infty} \left(\frac{1}{125}\right)^i$$

The summation above is an infinite decreasing geometric series, according to A.7 in the textbook:

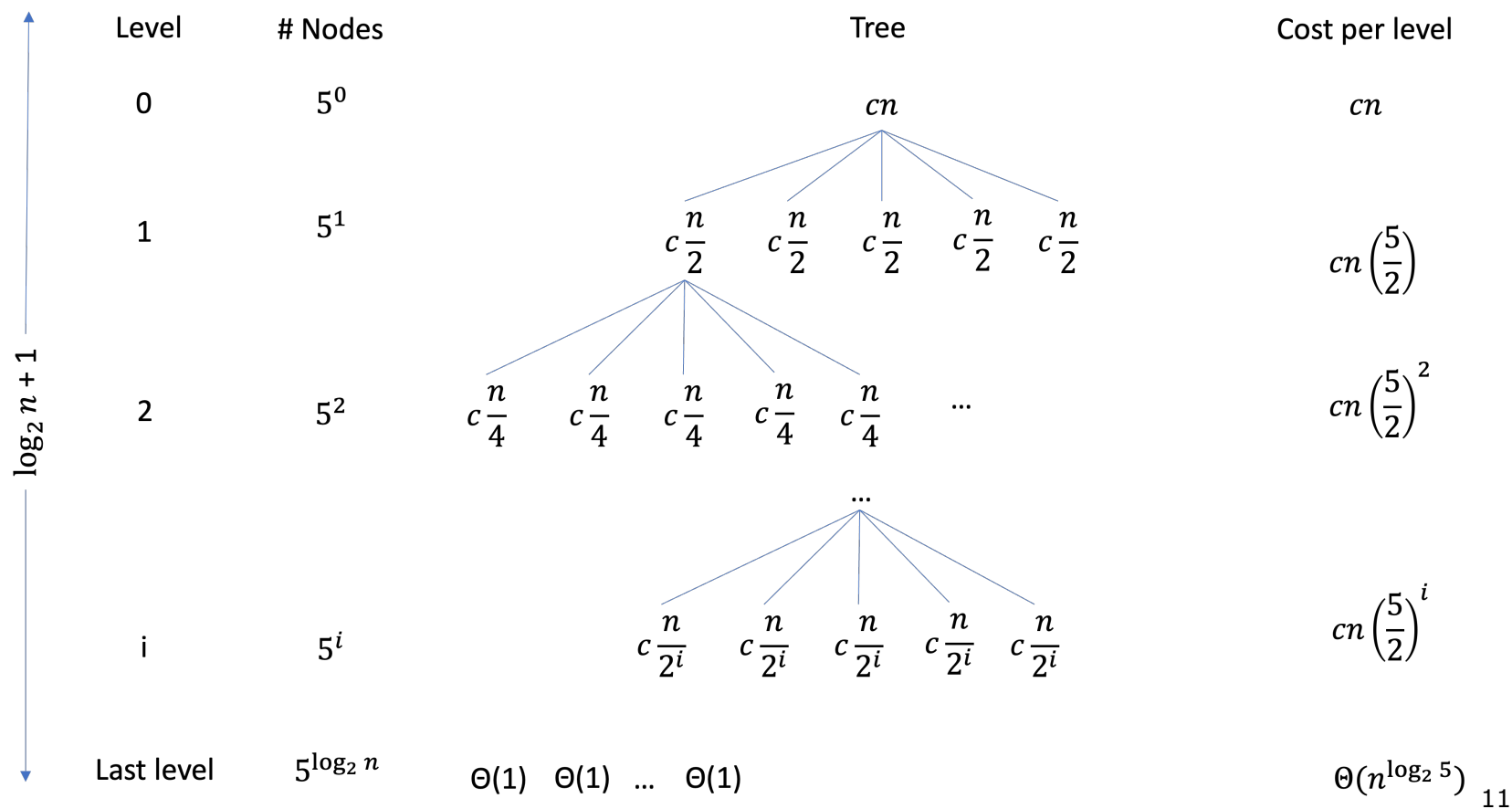
$$\sum_{k=0}^{\infty} (x)^k = \frac{1}{1-x}$$

, therefore, we have:

$$cn^3 \sum_{i=0}^{\infty} \left(\frac{1}{125}\right)^i = cn^3 \left(\frac{1}{1 - \frac{1}{125}} \right) = \left(\frac{125}{124}\right) cn^3 = O(n^3).$$

Assignment 1 – Exercise 2

$$T(n) = 5T(n/2) + \Theta(n).$$



Assignment 1 – Exercise 2

The total cost for the internal nodes + the last layer is given by:

$$\sum_{i=0}^{\log_2 n - 1} cn \left(\frac{5}{2}\right)^i + \Theta(n^{\log_2 5})$$

$$\sum_{i=0}^{\log_2 n - 1} cn \left(\frac{5}{2}\right)^i + \Theta(n^{\log_2 5})$$

$$= cn \frac{\left(\frac{5}{2}\right)^{\log_2 n} - 1}{\frac{5}{2} - 1} + \Theta(n^{\log_2 5})$$

$$< cn \left(\frac{5}{2}\right)^{\log_2 n} + \Theta(n^{\log_2 5})$$

$$= cn(5)^{\log_2 n} \left(\frac{1}{2}\right)^{\log_2 n} + \Theta(n^{\log_2 5})$$

$$= cn(5)^{\log_2 n} (2)^{-\log_2 n} + \Theta(n^{\log_2 5})$$

$$= cn(n)^{\log_2 5} (2)^{-\log_2 n} + \Theta(n^{\log_2 5})$$

$$= cn^{\log_2 5 + 1} (2)^{-\log_2 n} + \Theta(n^{\log_2 5})$$

$$= cn^{\log_2 5 + 1} (n)^{-\log_2 2} + \Theta(n^{\log_2 5})$$

$$= cn^{\log_2 5 + 1} (n)^{-1} + \Theta(n^{\log_2 5})$$

$$= cn^{\log_2 5} + \Theta(n^{\log_2 5}) = \Theta(n^{\log_2 5})$$

Assignment 1 – Exercise 3

a. $T(n) = 3T(n/9) + \sqrt{n} \lg^2 n.$ (15%)

The watershed function is: $n^{\log_b a} = n^{\log_9 3} = n^{\frac{1}{2}} = \sqrt{n}.$

The driver function $f(n)$ is $\sqrt{n} \lg^2 n.$

Therefore, case 2 applies with $k=2$. So, we have that $T(n) = \Theta(\sqrt{n} \lg^3 n).$

Assignment 1 – Exercise 3

b. $T(n) = 2T(n/4) + n^2$. (15%)

The watershed function is $n^{\log_b a} = n^{\log_4 2} = n^{\frac{1}{2}}$.

The driver function $f(n)$ is n^2 . So $f(n)$ is $\Omega(n^{\frac{1}{2}+\epsilon})$ for $\epsilon = \frac{3}{2}$.

For case 3 to apply, we need to check the regularity condition:

$af(n/b) \leq c(f(n))$ for some constant $c < 1$. Here $af(n/b) = \frac{n^2}{8}$.

Therefore, the regularity condition holds for $c = \frac{1}{8}$.